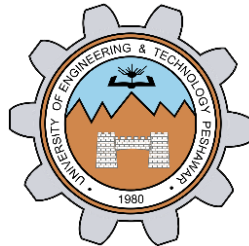


Lab # 07

Stacks Implementation



Spring 2026

Data Structures And Algorithms Lab

Submitted by: **M Mohsin Khan**

Registration No.: **24PWCSE2382**

Class Section: **C**

Submitted to:

Engr. Abdullah Hamid

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Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Task 01 Stack via Linked List Arrays:

```
#include <iostream>
using namespace std;
#define SIZE 100
struct Stack
{
    int arr[SIZE];
    int top;
};

void createStack(Stack *s)
{
    s->top = -1;
}

void push(Stack *s, int value)
{
    // if (s->top >= 0 || s->top <= SIZE)
    // invalid condition
    if (s->top <= SIZE - 1)
    {
        s->top++;
        s->arr[s->top] = value;
    }
    else
        cout << "Push function ----Stack overflow" << endl;
}

void pop(Stack *s)
{
    // s->arr[s->top] = 0;
    // s->top--;

    if (s->top == -1)
    {
        cout << "pop fucntion ----Stack underflow\n";
    }
}
```

```
        s->arr[s->top--];
    }
    int top(Stack *s)
    {
        if (s->top == -1)
        {
            cout << "top function ----Stack is empty! No top element." << endl;
            return -1;
        }
        return s->arr[s->top];
    }
    int size(Stack *s)
    {
        return s->top + 1;
    }
    void empty(Stack *s)
    {
        s->top = -1;
    }
    int main()
    {
        Stack s;
        createStack(&s);
        push(&s, 10);
        push(&s, 20);
        push(&s, 30);
        cout << "Top element: " << top(&s) << endl;
        empty(&s); // clears everything
        if (size(&s) == 0)
        {
            cout << "Stack is now empty." << endl;
        }
        return 0;
    }
}
```

Output:

```
PS D:\4th Semester\DSA\lab_07_stacks_via_LLArrays> .\a.exe
Top element: 30
Stack is now empty.
```

Task 02 Stack implementation via Link List through Pointers:

```
#include <iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

struct Stack
{
    Node *top;
    int count;
};

void createStack(Stack *s)
{
    s->top = nullptr;
    s->count = 0;
}

void push(Stack *s, int value)
{
    Node *newNode = new Node;
    newNode->data = value;
    newNode->next = s->top;
    s->top = newNode;
    s->count++;
}

void pop(Stack *s)
{
    if (s->top == nullptr)
    {
        cout << "Stack Underflow! Cannot pop" << endl;
        return;
    }
    Node *temp = s->top;
    s->top = s->top->next;
    delete temp;
    s->count--;
}
```

```
int top(Stack *s)
{
    if (s->top == nullptr)
    {
        cout << "Stack is empty! No top element." << endl;
        return -1;
    }
    return s->top->data;
}

int size(Stack *s)
{
    return s->count;
}

void empty(Stack *s)
{
    while (s->top != nullptr)
    {
        pop(s); // repeatedly delete nodes
    }
}

int main()
{
    Stack s;
    createStack(&s);

    push(&s, 100);
    push(&s, 200);
    push(&s, 300);
    cout << "Top element: " << top(&s) << endl;
    empty(&s); // clears the stack fully
    if (size(&s) == 0)
    {
        cout << "Stack is now empty." << endl;
    }
    return 0;
}
```

Output:

```
PS D:\4th Semester\DSA\lab_07_stacks> .\a.exe
Top element: 300
Stack is now empty.
```

RUBRICS (LABS/ PROJECTS):

Criteria & Point Assigned	Outstanding 4	Acceptable 3	Considerable 2	Below Expectations 1	Score
Valuing related to work, including responsibility, and professionalism.	Demonstrates excellent responsibility and professionalism in task management. Shows strong commitment to quality work, meeting the deadline, and comprehensive, well-organized effective documentation following all guidelines with exemplary attention to detail.	Demonstrates some responsibility and professionalism. Displays moderate commitment to quality work, meets the deadline and shows sufficient thoroughness in documentation, adhering to guidelines, with some room for enhancement.	Demonstrates limited effort in responsibility and professionalism. Misses the deadline with notable inconsistency in quality work and shows insufficient thoroughness or clarity in documentation, necessitating improvements.	Demonstrates a lack of responsibility and professionalism. Misses the deadline and lacks quality work. Documentation is incomplete, disorganized, or unclear.	
Ability to apply techniques, methods, and procedures effectively.	Executes lab tasks with exceptional skill to design/develop solution and shows proficiency in lab techniques and procedures.	Executes lab tasks with adequate skill to design/develop solution and shows proficiency in lab techniques and procedures.	Executes lab tasks with minimal skill to design/develop solution and shows moderate proficiency in lab techniques and procedures.	Executes lab tasks with limited skill to design/develop solution and lacks proficiency in lab techniques and procedures.	

Demonstrate understanding and relevant Knowledge.	Demonstrates exceptional understanding of concepts with clear explanation of investigation and analysis of the objectives and outcomes.	Demonstrates a good understanding of concepts with some explanations of investigation and analysis that align with the lab's objectives and outcomes.	Demonstrates a basic understanding of key concepts but lacks clarity or depth in explaining the investigation and analysis, with some inconsistencies.	Demonstrates limited understanding of concepts with unclear explanations of investigation and analysis that do not align with the lab's objectives and outcomes.	
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